OEM Reversing Hi line Camera retrofit. Needs coding. Yelland v02 – 2nd Aug 2024

https://www.tiguanforums.co.uk/threads/oem-reversing-hi-line-camera-retrofit-needs-coding.7470/?nested_view=1&sortby=oldest

Originally posted above in Nov 18, 2020 note this is what I did to install the camera not what is necessarily the best or correct way to do it! Just a record of what I did and there may be better way to do things! (This PDF has been updated to include other info and comments etc. from the post above as it developed).

After some research I found I needed a VW camera part number 5NA827566D which is the full line camera (i.e has moving steering lines while reversing).

I bought a genuine VW kit on EBay from vwmodul_de that had the camera with washer jet, loom and washer pipe, (and apparently the camera was 'precalibrated'). The kit also came with comprehensive connection instructions and coding detail but no 'installation instructions. Hence the original post I did.



You can also purchase separately:

Camera = VW pn 5NA827566D with washer jet. (Note this may not be calibrated but will work just the same but generates an error when scanned - see P13 of this document)

(There also seems to be lots of Chinese copies of this camera which I cannot vouch for. My kit was a genuine VW)

Washer pipe = VW pn 658 955 663 C

Loom (Video and connection) = VW Pn?
But search Aliexpress for "HighLine Rear View Camera Cable Tiguan" for example to find lots of alternatives. (Note the power supply wire in these variants tends to be white instead of red)

I did a test connection first and did the coding using OBDEleven Pro just to make sure everything worked before I did the proper installation. I got the details of what to code in the instructions with the kit and it wasn't complicated. Coding took less than 5 minutes.

Full installation was quite a lot of work but 'fun' (around 4 hours for me). It isn't that difficult, basic DIY skills, but some bits are fiddly and time consuming. I guess it would be a lot quicker the second time.

In the end this is what you get, (and you find all usage instructions in the VW owner's manual) Note I was sat on my drive and the car didn't move between pictures so you get an idea of the views.

a) Go into reverse and you get this default view, I had the steering wheel turned. The yellow lines showing where the car would go in reverse as you steer.



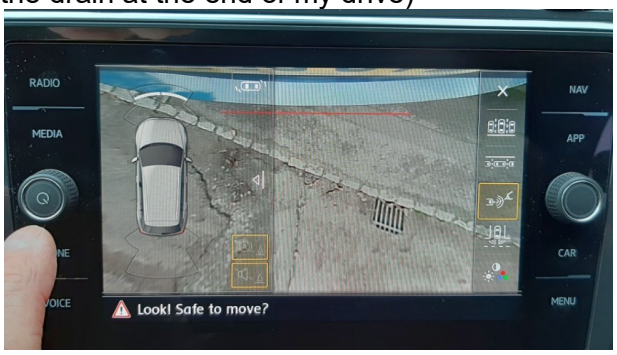
Note you get the sensor popup on the left and also the mute and disable Autobrake keys. You also get a button to remove the popup and a button to revert to the parkpilot view (like before you fitted the camera). You also get a set of buttons on the right to give different views as follows:

b) A fish eye view for maximum view to the sides and rear. The icon shows this as a view for reversing into a main road for example.

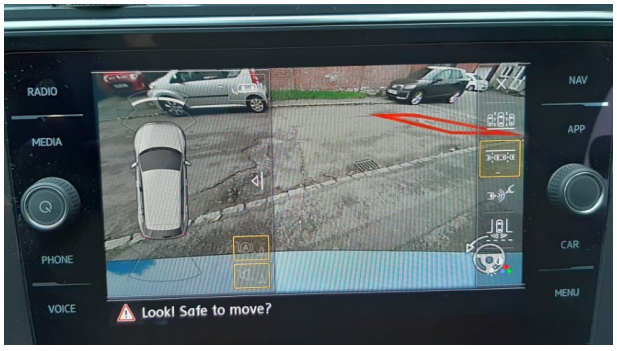


Compare picture a) above to get the 'extra wide view' effect

c) A straight down view useful for reversing up to a tow hitch etc. Or very close to a wall. (Or a view of the drain at the end of my drive)



d) You also get the 'parking assistance view', this seems to offer visual steering guidance parallel parking. The view from my drive doesn't show what it does really but the red square is where a parking bay could be and you essentially get steering guidance on the screen.

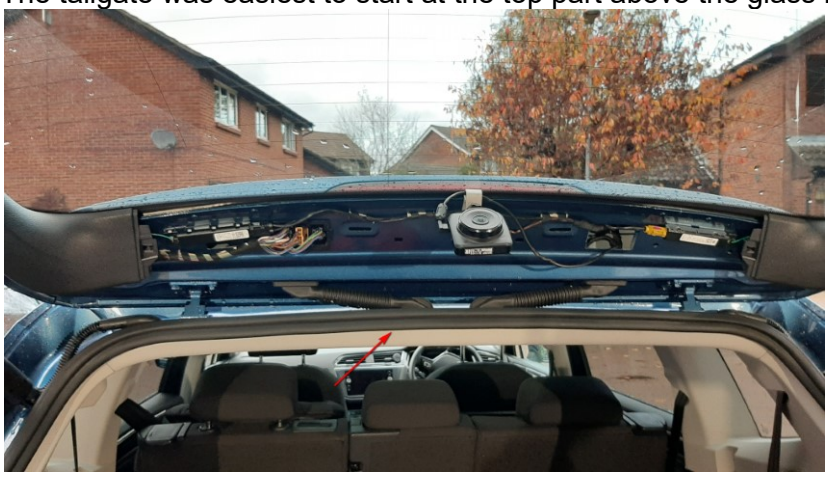


Fitting

(Tools and extras used are detailed at the end to the document P14)

1) I started at the camera end and wired forwards as you need to get the connectors through the boot lid grommet which is 'fiddly'. I took all the tailgate panels off, although some are quite tight. If you feel the plastic bending too much then pause to get a tool in and look around for where the clips are.

The tailgate was easiest to start at the top part above the glass next the hinges. This pulls off straight.



Then the two panels either side of the glass next. The top of these closest to the hinges pull out first - straight pull but the bottom is sort of twisted into the bottom panel. A lot of wiggling at the bottom worked for me. (Once it's out you see how the sides clip into the bottom panel for reassembly)

Then the bottom big panel of the tailgate. There are two Torx screws inside the flap where the warning triangle clips. Also one screw in the hand hold hole that you pull the lid down with.



I then started pushing the tools in around the bottom catch mechanism, moving around sideways to release the clips as you go. I also found that removing the access panels that enable you to change the bulbs in the rear hatch light assemblies helped see where the clips are.



Once the bottom was free and hanging I released the looms to the red and white light on the left side and the close button switch on the right. (Tricky as you have to release little clips in the connector).

Then pulled the rest of the clips off and fully removed the bottom panel.

2) Removed the existing boot lid opening switch and fitted the new switch with camera.
Bit awkward as it's above your head working. The old clips are fiddly. Look at the new camera clips to see how they work. There are 6 to release. I used some water pump pliers and small screwdrivers to release two side clips and an end clip at one end of the camera first then push camera through a little then do other end and fitted the camera. Be careful what orientation the new switch and camera is fitted.



3) Removed the high brake light and connected the washer pipe T piece supplied with the kit.
This is clipped in from the outside of the car. Just prise gently out

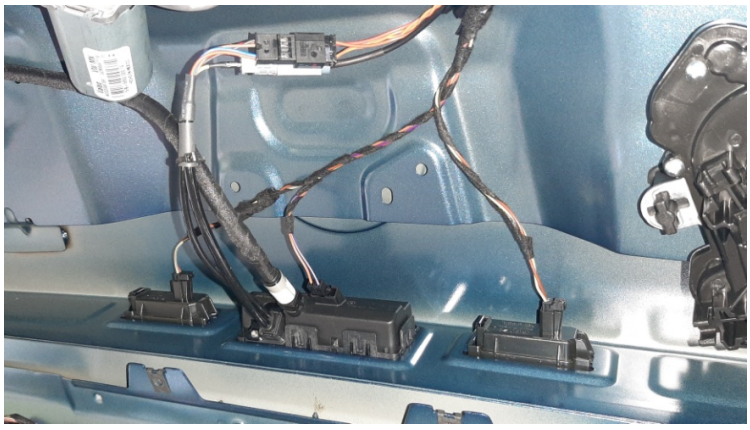




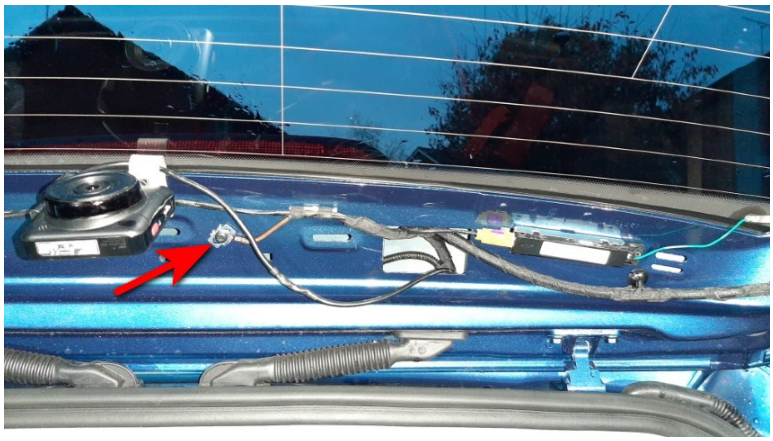
4) Routed the pipe down the right side of the lid and used existing clips to hold it in place.



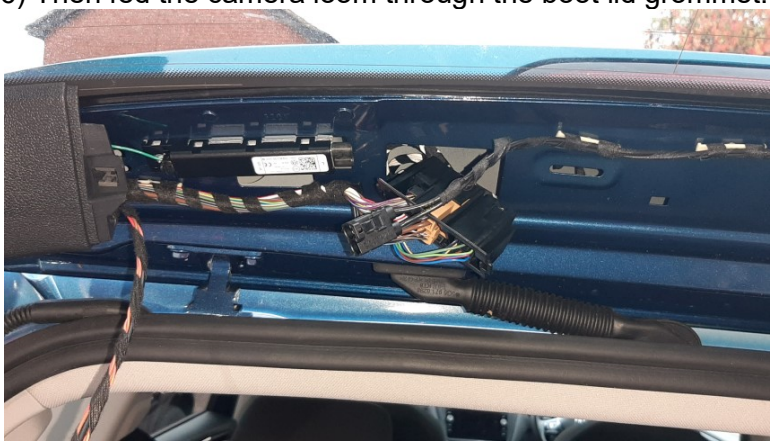
5) Connected the existing switch loom to the camera, connected wash pipe, connected the 2 new camera connectors and routed new loom down the left side of the lid (I guess it could go down the right side too)



There was a convenient earth point next to the hole for the rubber grommet to connect camera earth wire to.



6) Then fed the camera loom through the boot lid grommet. It's tricky!



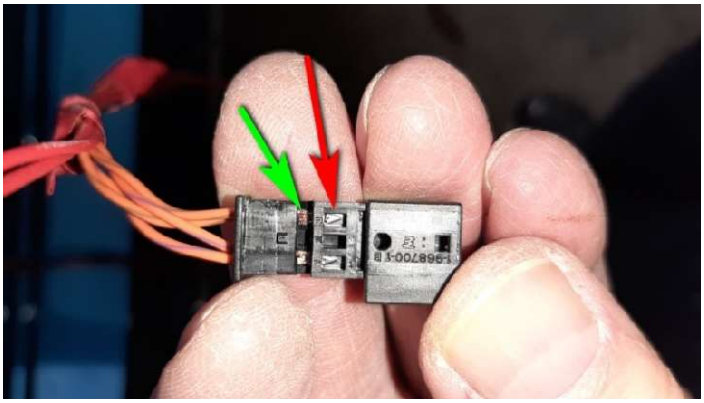
Release the roof lining - first separate the rubber seal from the roof lining by running a trim tool along the edge. Then open the headlining a little to look in. There are three clips into the roof. I used one of the trim tools to release the clips down to get access. Just take it easy so as not to crease the lining. I only opened the roof lining a bit to allow me to get my hands in to route the wire through the flexible grommet.



I found the 3 pin connector was a bit big to fit through the grommet so I ended up taking the connector housing apart to feed the wires through.

Remove the connection pins from the connector housing by pressing the springy metal 'barb'. If you wiggle the pin first in and out gently you will see it. Just press it down gently to allow the wire to come out. It's a 2 step process. Press - pull to come half way then press pull again. Note it is really a gentle pull if you have the barb pressed properly. No force else it will be damaged. Note which wire goes where in the connector.

I got there in the end being careful not to damage the crimps using tape to protect everything.



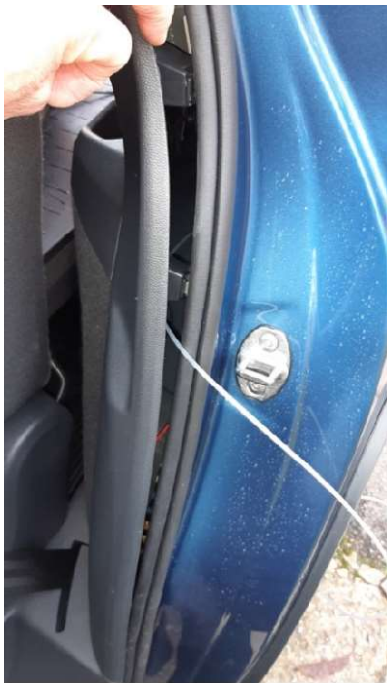


7) Releasing the side panel top edge to route the cable down to go through to the rear passenger door sill area.



8) Loosened the door sill rear, vertical wheel arch cover and front sill plastic trims and fed the wires through.

In terms of getting the loom from the boot area to the rear passenger door sill area I found that inserting the fishing line tool from front to rear from the rear door lock area as shown in the photo below and then pulling the wire forward worked well.



Then just ease the rear and front passenger side sills open to tuck the wires in.

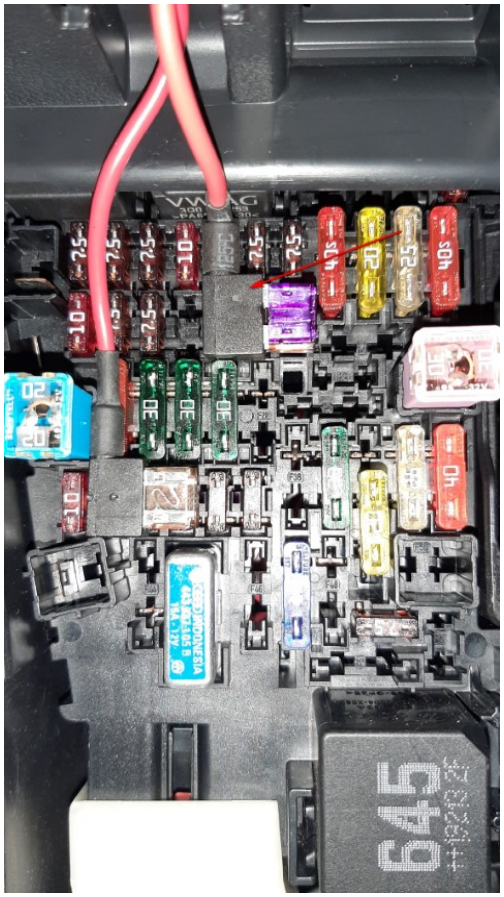


Then removed the passenger ‘airbag off’ switch panel on the end of the dashboard and fished the wires up vertically into that area.



9) Connected the red power wire to a piggy back fuse connector that is **permanently on** to the fuse box. I used a mini piggy back fuse connector to a permanently on connection in the fuse box. I fitted a 3 amp fuse (guessed that value). Right hand connection (Red arrowed) is the reversing camera (Left one is just my dash cam on a switched feed). Also see more details on P15.

Note: the camera wiring loom does have a terminal fitted so it can be inserted into the fuse box itself and a normal fuse then fitted. However this process is tricky and not something I wanted to do. So I used the piggy back instead.



10) Removed the DVD unit using some keys you can buy on ebay (see p14) and removed the rear connector to connect the 4 wires from the camera as per the very good/comprehensive instructions provided by the kit supplier I used.

Replace DVD unit

NOTE: When you remove the two 'add in wires' to the grey connector you hinge the side open as per kit suppliers instructions, it says: 'With the flat screwdriver, push the pins lock, and take the wires out'.

This is true but the 'lock' is a little springy metal 'barb'. If you wiggle the pin first in and out gently you will see it. Just press it down gently to allow the wire to come out. I recall it's a 2 step process. Press - pull to come half way then press pull again. Note it is really a gentle pull if you have the barb pressed properly. No force else it will be damaged.

All done fitting wise. Only coding to do!

Coding

Manual coding

I have an OBDEleven Next Generation Pro. The coding was simple adaption and long coding all detailed in the kit supplier instructions. No credits required. (VCDS can also be used)

This is my more wordy and slightly corrected long coding from what I found on my car when I did my installation. I guess names change with newer / older systems and software versions etc.

Probably good to do a back up of the settings in case you get lost - or be very meticulous and write everything down as you go on paper / or take screen shots.

Open Bonnet. (**Essential**)

Ignition on and OBD Eleven plugged in and App connected.

Goto Module 19 Gateway
click on Adaption – then Installation list – then add 'CAMERA SYSTEM REAR VIEW' (Mine was originally not selected)

Goto to Module 5F Multimedia
Click on Adaption – then Car_function_List_BAP_Gen2 then VPS_oxoB set to ACTIVATED (Mine was originally not activated)

Then Car_function_List_BAP_Gen2 then VPS_oxoB_msg_bus set to DATABUS INFOTAINMENT (Note was already set to this in my car)

Goto 76 Parking / Steering assist (instructions said module 10 or 76. 10 didn't exist for me)
click on Long coding – enable the Camera Type – Rear view camera. (was set to nothing for me)

Now restart the Multimedia unit to load the new settings. I did this by switching ignition off and unplugging power to the unit. But apparently you can do a manual reset by holding down the left hand knob until the logo comes back on the screen and let go.

App coding

Since I did my camera there is now an OBDEleven 'Rear view camera high' retrofit app. The app costs 100 credits (17.99 Euros currently)

(Note the retrofit app instructions in OBDEleven aslo include some connection details).

All done!

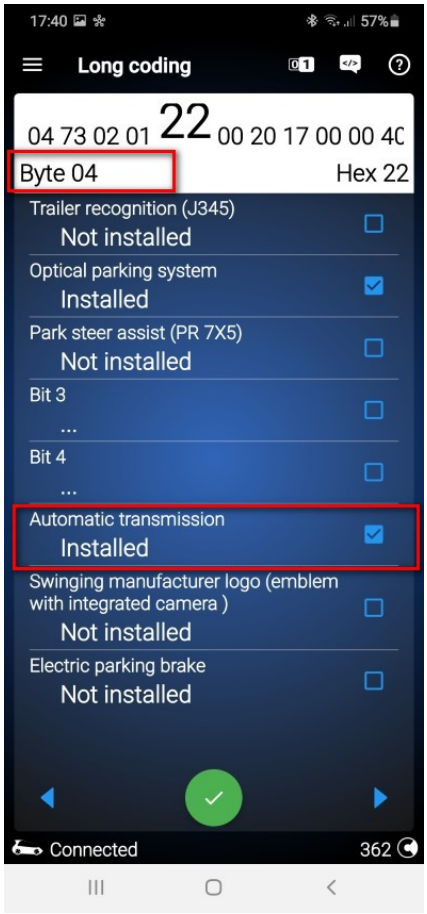
Module 6c Rear View Camera possible errors after installation

If you rescan for errors etc after installation you may find these.

1) Manual Gearbox

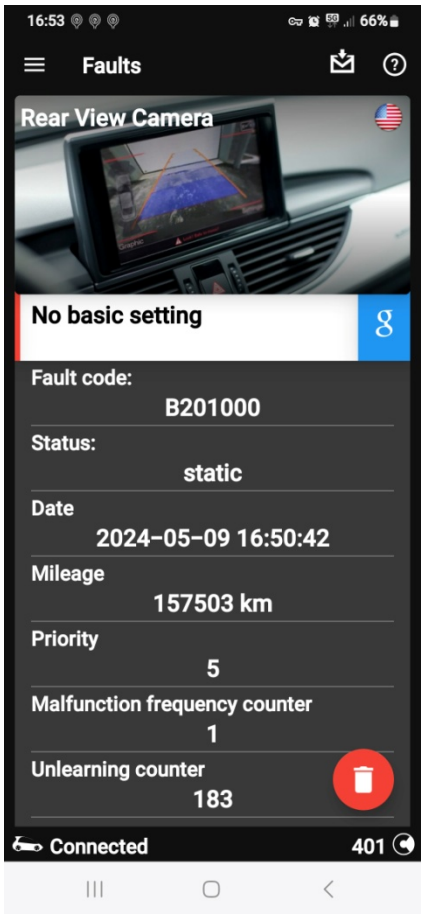
Note: The standard coding in for my camera module 6c assumes a DSG gearbox (not manual)

This can give an fault code when reading with OBDEleven or VCDS. You can just ignore the code but if required this can be corrected by going to the module 6c long coding and deselect Byte 04 Bit 5 'Automatic transmission' (highlighted below)

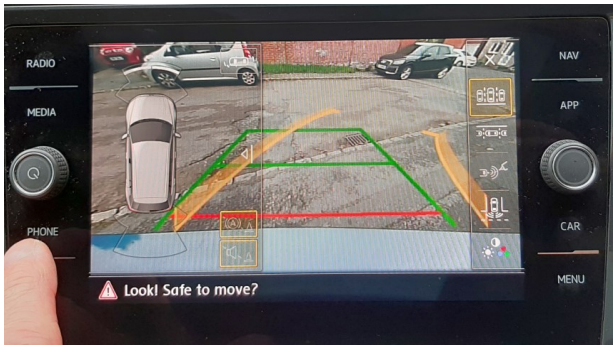


2) No Basic Setting

Note: The camera I purchased had ‘calibration data’ already stored in it. If the camera doesn’t have this data then module 6c can give an error ‘no basic setting’ when reading with OBDEleven or VCDS. This shows the camera is not calibrated but everything still seems to function ok if you just ignore it.



For reference if I reversed up to a brick wall for example then on my car the reversing sensors start beeping when the outer green line below approximately touches the wall. The continuous tone is with the red line approximately touching the wall.



Tools and 'extras' I used:

1) Trim removal tools
If you search say EBay for “11PC TRIM REMOVAL PRY TOOL KIT” you will find loads of options. e.g.



2) DVD unit removal in the glove box
You need two keys to remove the dvd unit. Again loads on ebay search 'PC5-132 key'. e.g.



3) Fishing wires through
Very useful for generally fishing wires though the car. Its a semi stiff plastic line. Loads on ebay – search Nylon Draw Tape (or available from Toolstation).



3) Piggy back fuse
When you connect to the main fuse board behind glove box (see post above) I used a piggy back fuse holder. They come in two sizes. Mini and Standard. Mini is the one I used as it's the main type in my car. EBay search "piggy back mini fuse holder"



First identify an existing fuse on the fuse board that is permanently 12v dc (not switched by the ignition). Then remove the old fuse. In the picture above you plug the two metal pins into the old fuse position and place the removed fuse into the bottom slot. The top slot is for the new fuse to the camera. Note in the picture above the piggy back fuse holder should be orientated so that the left metal pin goes to the live 12v of the fuse panel.

(See the fuse picture in P9 above to get the fuse location and orientation I used).

You also need a mini fuse. Some suppliers of the fuse holder also can provide a fuse with the holder just look for the correct value.

5) Crimping tool
I also used a crimping tool to connect the Piggy back fuse connector wire to the reversing camera positive feed wire, (or just cut it off and solder, or even choc block I guess, etc.)